# Jasmer Singh Sanjotra

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## Education

**Indian Institute of Technology Indore**, B.Tech in Electrical Engineering 2022 – 2026

- CGPA: 9.05/10

## Experience

**B.Tech Thesis: Causal Inference in Reinforcement Learning** Aug 2025 – Present

- Developed novel causal variants of policy gradient algorithms (**REINFORCE, PPO, and A2C**) by introducing **causal-aware action-independent baselines** to mitigate variance and bias.
- Theoretically derived **tighter regret bounds** and empirically validated findings on custom **Structural Causal Models (SCMs)** and high-dimensional **MuJoCo** environments.
- **Recognition:** First-authored research extension currently **under review for ICML 2026**.

**Teaching Assistant**, IIT Indore – Indore, IN Aug 2025 – Dec 2025

- Selected by Prof. Dibbendu Roy as a TA for EE319, a core course covering the mathematical foundations of communication systems for 40+ students.
- Conceptualized rigorous problem sets and quizzes focusing on **Elementary Measure Theory, Markov Chains, and Markovian Queueing Systems**.
- Conducted regular technical tutorials to facilitate discussions and explains concepts and questions.

**Developer Intern**, Samsung R&D Institute India – Bengaluru, IN May 2025 – July 2025

- **Engineered** novel deep learning frameworks for the **6G Standards Team**, focusing on high-fidelity CSI-RS (Channel State Information Reference Signal) recovery.
- **Designed** customized UNet-inspired models for spectral super-resolution, achieving a state-of-the-art Normalized Mean Squared Error (NMSE) of $10^{-3}$ on sparse, complex-valued 2D-FFT signals.
- **Patent & Recognition:** Authored an innovation currently under **US Patent filing**; secured an **Advanced Developer** return offer, awarded to top-performing interns for exceptional technical contributions.

**Research Intern**, Prof. M. Tanveer | OPTIMAL Research Group, IIT Indore – IN May 2024 – Present

- Developed LSTMSE-Net, an audio-visual speech enhancement model to isolate and enhance speaker audio in noisy environments.
- Engineered a temporal feature extraction pipeline using **RNN and LSTM** units to jointly model audio-visual dependencies.
- Achieved a **3× reduction** in inference time compared to the baseline model with improvements in speech quality.
- Original paper accepted in InterspeechW 2024.

**Research Intern**, Prof. Nagendra Kumar | LIPG, IIT Indore – IN May 2023 – Present

- Developed novel U-Net based deep learning models for liver tumor segmentation, handling data pre-processing and **creating custom callbacks, metrics, and loss functions**.
- Implemented advanced architectures like **Squeeze-and-Excitation Networks and Atrous Spatial Pyramid Pooling**, achieving a state-of-the-art accuracy of **98%**.
- Co-authored a research paper detailing the methods and results. Published in Elsevier Biomedical Signal Processing and Control.

**Research Contributor**, Dr. Debesh Jha | Northwestern University – IL, USA May 2024 – July 2024

- Implemented various liver tumor segmentation models including **DeepLabv3+, UNet, and HiFormer-L** on the LiTS dataset using PyTorch and TensorFlow.
- Engineered **custom, modular PyTorch data loaders** and transformation pipelines for the LITS dataset.
- Source code available at GitHub/TheAlphaJas/Liver-Tumor-Seg-Implements.

## Publications

**LSTMSE-Net: Long Short Term Speech Enhancement Network for Audio-visual Speech Enhancement** — Sep 2024

Arnav Jain*, ***Jasmer S. Sanjotra****, Harshvardhan Choudhary, Krish Agrawal, Rupal Shah, Rohan Jha, MD Sajid, Amir Hussain, M. Tanveer

10.21437/AVSEC.2024-8
*3rd COG-MHEAR Workshop on Audio-Visual Speech Enhancement (AVSEC)*, ISCA

**Ethical framework for responsible foundational models in medical imaging** — 2025

Debesh Jha, Gorkem Durak, Abhijit Das, ***Jasmer Sanjotra***, Onkar Susladkar, Suramyaa Sarkar, Ashish Rauniyar, Nikhil Kumar Tomar, Linkai Peng, Sirui Li, Koushik Biswas, Ertugrul Aktas, Elif Keles, Matthew Antalek, Zheyuan Zhang, Bin Wang, Xin Zhu, Hongyi Pan, Deniz Seyithanoglu, Alpay Medetalibeyoglu, Vanshali Sharma, Vedat Cicek, Amir A. Rahsepar, Rutger Hendrix, A. Enis Cetin, Bulent Aydogan, Mohamed Abazeed, Frank H. Miller, Rajesh N. Keswani, Hatice Savas, Sachin Jambawalikar, Daniela P. Ladner, Amir A. Borhani, Concetto Spampinato, Michael B. Wallace, Ulas Bagci

10.3389/fmed.2025.1544501
*Frontiers in Medicine*, Volume 12, 2025

**MNet-SAt: A Multiscale Network with Spatial-enhanced Attention for Segmentation of Polyps in Colonoscopy** — 2025

Chandravardhan Singh Raghaw, Aryan Yadav, ***Jasmer Singh Sanjotra***, Shalini Dangi, Nagendra Kumar

10.1016/j.bspc.2024.107363
*Biomedical Signal Processing and Control*, Volume 102, 2025

**T-MPEDNet: Unveiling the Synergy of Transformer-aware Multiscale Progressive Encoder-Decoder Network with Feature Recalibration for Tumor and Liver Segmentation** — 2025

Chandravardhan Singh Raghaw, ***Jasmer Singh Sanjotra***, Mohammad Zia Ur Rehman, Shubhi Bansal, Shahid Shafi Dar, Nagendra Kumar

10.1016/j.bspc.2025.108225
*Biomedical Signal Processing and Control*, Volume 110, 2025

## Projects

**rawML: Autograd Engine & Neural Framework from Scratch** — Oct 2024

- Engineered a lightweight, NumPy-based deep learning library featuring a **custom Autograd engine** (jTensor) with dynamic computational graph construction and gradient tracking.
- Implemented vectorized modular layers, loss functions, and optimizers (SGD, Adam), enabling training of multi-layer perceptrons without high-level framework dependencies.

**Memory-Augmented RL for Stochastic Word-Games** — Aug 2024

- Designed a **Deep Q-Network (DQN)** agent integrated with an **RNN (LSTM/GRU)** architecture to handle the partially observable state space of the Hangman game.
- Utilized **Experience Replay and Target Networks** to stabilize training, achieving an 85%+ win rate across a vocabulary of 50k+ words.

## Technical Skills

**Languages:** C++, Python, MATLAB

**Frameworks & Tools:** Git, PyTorch, Tensorflow, OpenCV, NumPy, backtesting.py, UnTrade SDK, Pennylane

**Operating Systems:** Windows, Linux

**Other Skills:** Competitive Programming, Data Structures and Algorithms, Object Oriented Programming, Probability and Statistics

## Achievements

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| **Received IEEE SPS ME-UYR Grant**, In collaboration with UoG, Netherlands | 2025 |
| **Expert** on Codeforces, demonstrating strong coding and reasoning ability | handle |
| **Selected for Amazon ML Summer School**, Amazon MLSS | 2024 |
| **All India Rank 3002** (out of 200k+ aspirants), JEE Advanced 2022 | 2022 |
| **99.1%ile** (out of 800k+ aspirants), JEE Mains 2022 | 2022 |
| **IOQM Certificate of Merit**, Indian Olympiad Qualifier in Mathematics | 2022 |